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Reservoir Boost AI: Redefining Waterflood Management in Mature Assets

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Abstract

This paper describes the experience gained from the development and application of data-driven modeling techniques to enhance the efficiency of waterflood operations in several mature fields in Block 61, Ecuador. A modeling application that integrates data analytics, machine learning (ML) and physics-based reservoir modeling has been prototyped since 2022 as part of a wider digital transformation initiative. The goal is to significantly enhance decision-making efficiency, reduce analysis time, and support proactive field management strategies. The application is designed to help multidisciplinary teams by simplifying complex reservoir behavior and fostering accessibility for both experts and non-experts in secondary recovery.

After over 50 years of production by pressure depletion, these partially exploited, logistically challenging, oilfields have been actively developed since 2017 using horizontal well technology and waterflooding. Initially water injection allocation was decided using simple analytical methods with grossly simplifying assumptions and spreadsheet calculations that were found to be ineffective and tedious as complexities in the management of the waterfloods began to unfold. After exploring various alternatives and experimental prototyping, including pure AI/ML, numerical reservoir simulation, surrogate modeling, management decided to develop a hybrid AI-Physics application based on capacitance-resistance and fractional flow theories to replace the spreadsheet. The new smart injection allocation system supports seamless data ingestion from a corporate database that stores daily reconciled production and injection data. An interactive web-based interface empowers users to evaluate multiple optimization scenarios, perform what-ifs with short-term forecasting capability to support decision-making.

Leveraging several complementary digital technologies, the system has been instrumental in offering new insights, increasing efficiency and recovery through informed decision-making. Analysis time has reduced from several hours to a few hours, but more importantly, blind testing has proven the application to be remarkably dependable. Since its deployment in September 2023, the application has contributed to an estimated incremental cumulative oil production of over 128 KBO across the fields analyzed, with an economic impact of 3.81 MMUSD to date. In addition to operational gains, the system has improved cross-team collaboration, enabling faster, more consistent responses to field conditions. Its deployment

also contributes to sustainability goals by minimizing CO₂ emission through decreased need for on-site monitoring.

Introduction

Block 61, operated by the Shaya Consortium, stands as one of Ecuador's most strategically important and productive oil assets, contributing approximately 19% of the country's total crude oil output. Discovered in 1970 and brought into production in 1975, the block currently maintains an average daily production of 71,000 barrels of oil per day (BOPD), with historical peaks reaching up to 85,000 BOPD. Spanning an area of 783 km², Block 61 is geographically divided into three operational zones north, central, and south and encompasses 17 producing fields and four principal reservoirs: Basal Tena, Napo U, Napo T, and Hollín.

The Brown field located in the northern sector of the Oriente Basin has been producing since 1972 and has surpassed 787 million barrels of cumulative oil production. In response to the natural decline associated with primary recovery, secondary recovery efforts began in 2017 with the implementation of a waterflood pilot in the Culebra-Yulebra area, targeting the Lower U Sandstone. Since then, water injections have progressively expanded to additional reservoir units such as the T Inferior, Basal Tena, and T Superior formations. Currently, secondary recovery contributes approximately 40% of the field's total daily production.

Initial water injection strategies in the asset primarily relied on deviated wells for both injectors and producers, which allowed for manageable control of sweep patterns and pressure support. As reservoir knowledge evolved and the need for improved reservoir contact became evident particularly in laterally extensive or geologically heterogeneous zones—horizontal wells were progressively incorporated, especially on the production side. These horizontal producers offered the potential for enhanced areal coverage, improved drainage, and delayed water breakthrough in some scenarios. However, their deployment also introduced significant technical and operational complexity, particularly in the context of waterflood management.

To navigate challenges and develop a more reliable understanding of reservoir behavior under such complex conditions, a comprehensive suite of reservoir characterization and diagnostic tools has been deployed. This includes geological modeling to frame connectivity hypotheses, interference testing to confirm pressure communication paths, dynamic reservoir simulation to evaluate water movement under varying operational constraints, and chemical tracer analysis to confirm and quantify flow connections. Complementing these studies simple analytical methods assuming fixed-patterns, first line producers around injection wells, had been used for operational purposes to recommend set points for injectors and producers. A spreadsheet tool was used to help balance the waterflood. However, the fixed-pattern assumption was very restrictive considering that reservoir simulation showed connectivity beyond the first line producers. The spreadsheet tool could process only one pattern at a time and was not able to consider pattern interactions. The method did not support the modeling of water production and production forecasting, limiting its use considerably.

As part of a broader digital transformation strategy the need for a smart injection allocation and pattern balancing system was identified as a priority. An innovation project was undertaken in late 2021 to deliver this need. Following a design thinking approach, existing solutions were reviewed, innovative ideas were prototyped and after some experimentation a deployable prototype was developed and examined in a field pilot (Khataniar et al., 2023). The capacitance-resistance modeling (CRM) engine was found to fulfill most of the needs of the waterflood monitoring reservoir engineers involved in the project over an implementation of the VAR-X (variable auto-regression with exogenous variables) method. Results were verified against numerical simulation results for both known problems and field models (Grijalva et al., 2023).

Based on the feedback from using the deployable prototype, the next phase of the project concentrated on producing a deployable solution that would apply to other fields and other teams than the field on which the solution was prototyped. Several enhancements were made on data automation, engine features

and performance and usability improvements. This paper describes a proprietary deployable solution, AI-Waterflood Insights (AWI), and two application examples to illustrate its utility. The solution enables real-time monitoring, improves understanding of injection-production dynamics, and supports prompt operational decision-making to maximize recovery efficiency. Since the introduction of AWI an estimated incremental production of about 128,000 BO has been achieved.

The application of this AWI in two real-world case studies will be presented in this work, highlighting its effectiveness in supporting data-driven reservoir management and improving the performance of mature waterflood projects.

Challenges

The horizontal well architecture, with extended laterals traversing multiple stratigraphic layers and facies, significantly complicates the control of water injection fronts. Ensuring uniform displacement along the entire horizontal section is inherently challenging, as variations in permeability, reservoir pressure, and saturation across the lateral often result in preferential flow paths. These conditions can lead to early water breakthrough in high-permeability zones while leaving large portions of the reservoir under-swept. Controlling the onset and progression of water breakthrough across the full lateral extension of horizontal wells requires a nuanced understanding of reservoir heterogeneity and inter-well dynamics, which is rarely available in real time.

Further complexity arises from injection patterns that combine deviated and horizontal wells. These mixed well geometries often disrupt conventional five- or nine-spot patterns, resulting in irregular injector-producer alignments and complex pressure fields. The variability in injector-producer spacing, orientation, and vertical depth makes it difficult to apply standard pattern balancing techniques and leads to asymmetric sweep and suboptimal volumetric efficiency. In some cases, this mismatch can result in overlapping drainage areas or dead zones within the reservoir.

In addition, surface constraints and logistical challenges compound the problem. The field's geographical dispersion and the significant distance between certain well pads and processing or injection stations limit the flexibility to optimize injection rates and pressures. Infrastructure bottlenecks such as limited availability of water lines, pump capacity, or injection points can delay or restrict the implementation of corrective actions, even when subsurface diagnostics clearly indicate the need for adjustment. These surface limitations often force operators to work with suboptimal operating envelopes, reducing the effectiveness of waterflood strategies and delaying response times to breakthrough events or injector realignment needs.

Deployable Solution

In this section, the enhancements made to the deployable prototype are presented. Description of the deployable prototype engine based on CRM (Yousef et al., 2006; Sayarpour et al., 2006, Cao et al., 2015) has been presented earlier (Grijalva, et al., 2023, Elfeel, et al., 2024). Grijalva et al. reported difficulties related to training the data-driven model to producers' liquid rate and water cut for some wells were highlighted. They recommended that incorporation of bottom-hole pressure could improve the fit quality. Elfeel et al. described parameter reduction techniques by assuming a relationship of the time constant to distance and computation of allocation factors from connectivity weights.

Inclusion of bottom-hole pressure

Majority of the wells in Block 61 are producing using ESP and pump-intake pressure (PIP) is a readily available parameter. The PIP is read from the database and processed to provide BHP input to the engine.

Regression algorithm performance

A significant amount of effort was spent in investigating different optimization routines to select defaults that would be suitable for deployment. Field engineers are not expected to understand optimizer strengths and weaknesses. A matrix formulation (Holanda et al.) was adopted and the code optimized for performance. The use of daily production data is a boon on the one hand but can increase demand for computational power. Parallel computation has been implemented to allow modeling of entire fields in real time.

Enhanced water-cut modeling

The Koval model used to fit water production is sensitive to observed data. Unfortunately, water cut samples can be quite noisy and hard to separate the signal from the noise. Some wells in Block 61 show a flat water-cut, usually less than 10%, profile over time. This is usually water from the transition zone or from aquifer encroachment into the injector pattern prior to waterflooding. Some modifications were done to the Koval model fitting algorithm and additional parameters were introduced to improve the fit quality.

Aquifer influx

Some of the fields in Block 61 have evidence of influx from small aquifers. A pseudo-injector modeling option has been included in the deployable solution.

Database connectivity

In the deployable prototype, input data and output results were managed as flat files in csv format. The deployable solution has a much more robust way of handling data and results with direct connectivity to a corporate database hosted on a commercial platform.

One of the main features of the workflow is the automation of data population and updating to bring all the information required to set up simulation cases. The workflow connects directly to the corporate database and brings all the data required. The user can select the necessary filters and grouping according to the hierarchy configured in the database to select only the necessary data, filters like Field name, Reservoir Name, and Zone name are currently used in this step of the workflow to create the main dataset (Figure 1).

Dataset Builder
Build datasets for injector wells, producer wells, well locations, and create a main dataset

Buttons: Create New Dataset, Clone Dataset, Modify Dataset, Update Dataset

Select a Dataset
AUCA_Central_UI (5293322912081634)

Data Source
Database

Database
Main Database

Add Field Filter (Optional)
AUCA CENTRAL X Select a field

Add Reservoir Filter (Optional)
UI X Select a reservoir

Add Zone Filter (Optional)
CENTRO X Select a zone

Dataset Name
AUCA_Central_UI (5293322912081634)

The main dataset will combine all selected columns from Injector Wells, Producer Wells, and Well Locations datasets.

Figure 1—Dataset setup and creation module.

Data processing

Digital data from sensors is available at a much higher frequency than the smallest interval (one day) design specification of the deployable solution. As such the averaging process can introduce noise that affects the results obtained. Sometimes these effects are not noticed till the training process is underway. Hence, a feature has been introduced that offers various algorithms for denoising input data. A machine learning-based denoising module enhances signal clarity,

An important aspect of dataset creation is data quality check to ensure the correct data is being compiled to minimize the uncertainty of the simulation results. The quality check analysis is shown below to give the user an insight into different statistics regarding the data to determine if further data validation is necessary. Also, the Data Visualizer incorporated in the workflow helped visualize the arrangement of different plots to make it easier for the user to determine the quality and validate the input data coming from the main database.

One of the main comparisons for data checking and quality control is the analysis of allocated production data versus well testing data. This comparison provides valuable insight into the degree of certainty with which the production information reflects the actual reservoir response and conditions. In the workflow, daily reported production data is used to train the model and detect patterns in reservoir behavior. However, this data may sometimes deviate from actual conditions due to measurement limitations, allocation assumptions, or operational constraints. To validate the accuracy and consistency of the daily data, Vx multiphase flow meters are used as a key reference. These meters provide more reliable real-time production measurements, allowing the team to verify whether the daily allocated data accurately represents field conditions. By comparing both sources, discrepancies can be identified, and corrective adjustments can be made to improve the quality of the dataset used in the modeling process, ultimately increasing confidence in the model's outputs and its representation of reservoir performance.



Figure 2—Dataset quality check module

User Experience

Starting with automated ingestion of production, injection, and pressure data, the application performs quality checks and generates interactive visualizations to validate data integrity. A machine learning-based denoising module enhances signal clarity, while customizable model configurations allow targeted training of the capacitance- resistance model. This process of ingestion and data quality check requires just a few minutes to process daily data of 10s of wells for up to 8 years at a time representing a significant improvement compared to the traditional time-consuming manual data preparation performed by analysts.

The core output dynamic injector–producer allocation factors feed directly into a PFM module. This enables engineers to identify sweep inefficiencies, rebalance patterns, and test improved injection strategies. The same engine supports production forecasting and what-if scenario analysis, helping teams assess the impact of operational decisions before implementation.

In summary, the solution offers end-to-end capabilities: from automated data preparation and quality control to advanced connectivity modeling, optimization, and forecasting empowering engineers with a scalable, fast, and collaborative application to drive better reservoir management and recovery strategies.

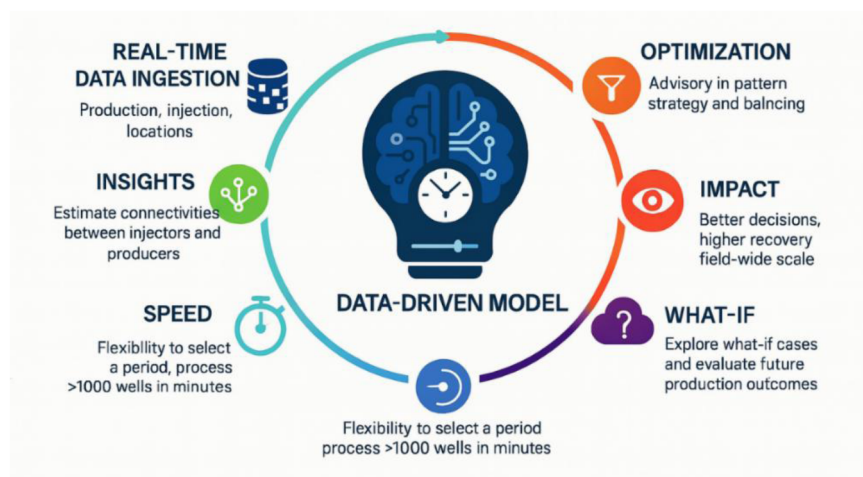


Figure 3—Deployable Solution Features

The solution has been successfully deployed in real field environments, enabling engineers to analyze, optimize, and forecast waterflooding performance with minimal manual effort. Its workflow combines robust data automation with advanced simulation and optimization capabilities.

Results

Since the implementation of the application, its use has been expanded across the entire Block 61. In this work, two representative case studies will be presented. These examples demonstrate the application's effectiveness in supporting data-driven decision-making and improving waterflood management.

Study 1

Problem Statement. This case study analyzes the performance of two adjacent injector-based waterflood patterns involving nine producing wells. The map below shows arbitrary pattern definition as used by the legacy spreadsheet tool (Figure 4).

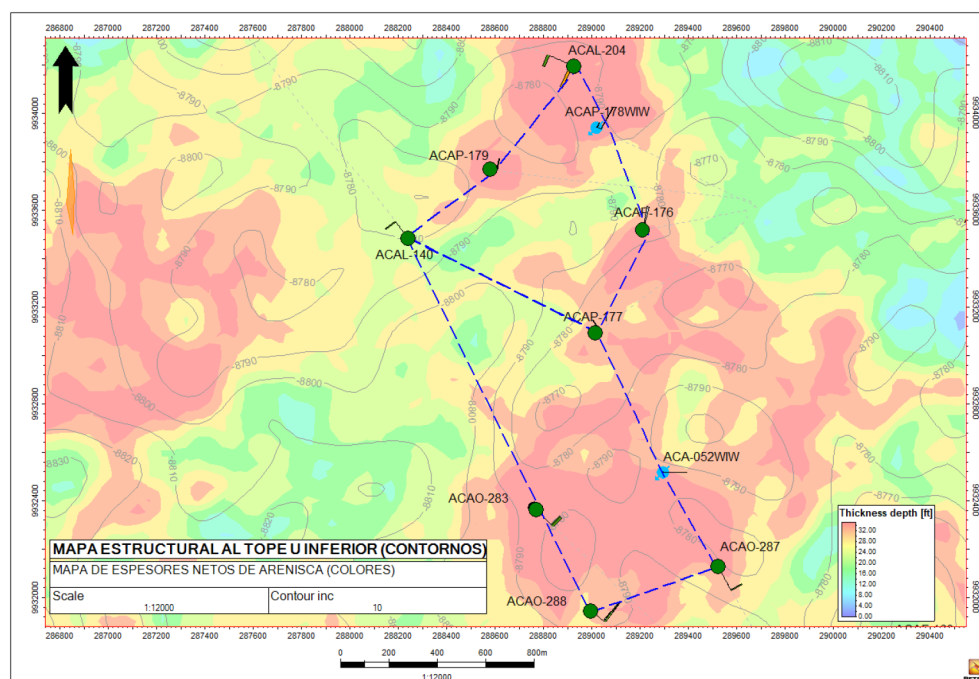


Figure 4—Location map of wells within the study 1 area in 2022 to 2023.

The oil production rate for injector well ACAP-178WIW pattern began to decline in early 2022 (Figure 5). There was uncertainty about how both patterns interacted with each other beyond the first line producers to the injectors since both injectors had stable injection at the time as shown in the Figure 5.

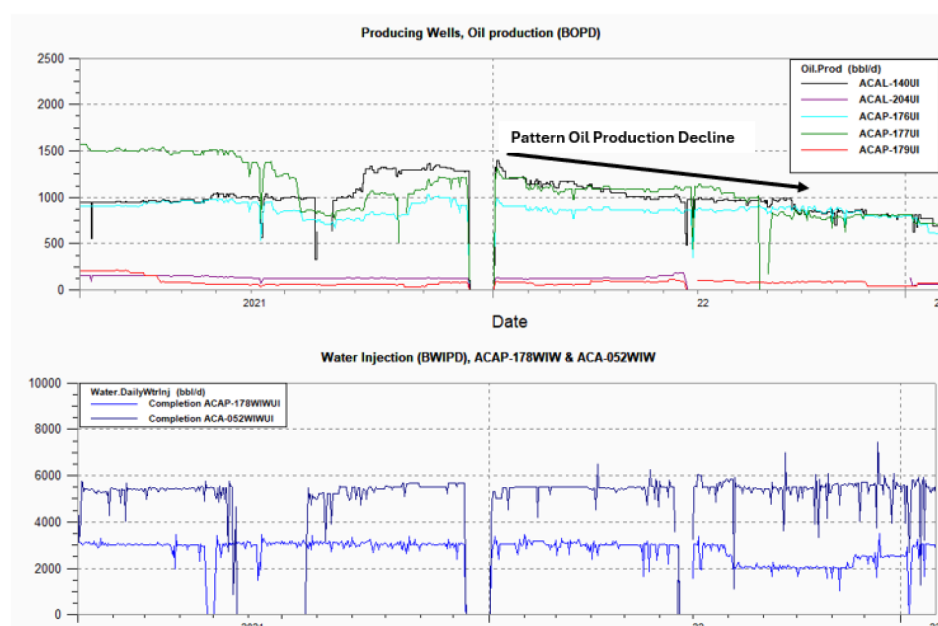


Figure 5—Wells' Production and Injection Behavior

A comprehensive diagnostic study was undertaken in the area to understand the reasons behind the reduced reservoir response, identify inefficiencies in the current injection distribution, and explore how to better utilize the available injection capacity to enhance oil recovery.

One of the main challenges in this area is the presence of overlapping injection patterns, where multiple producers are supported by shared injectors. This configuration complicates the interpretation of inter-

well connectivity and makes it difficult to attribute production behavior to specific injectors. Additionally, the geological complexity of the reservoir is characterized by lateral facies variations, discontinuous sand bodies, and heterogeneities in permeability and saturation further obscures the understanding of fluid movement and sweep efficiency.

Modeling. To address these issues, AWI solution was applied to evaluate connectivity, quantify injector contribution, and simulate optimized water distribution scenarios. The model includes the nine producers and two injectors, in which this study focuses in the upper pattern containing five producers – ACAL-204, ACAP-179, ACAL-140, ACAP-176, ACAP-177- and their relationship and influence with the two injectors – ACAP-178WIW, ACA-052WIW-. Once the data preparation was concluded, the model was trained with all the historical data since the start of the waterflooding injection project in both patterns in July 2018. Since both injector wells started injections around the same time it was important to understand the relationship and influence each injector had with each well in both patterns to have a proper balance of energy. The match was done using the entire production and injection data since the beginning of the waterflood water injection in these two patterns since July 2018. The period selected for model training is critical, as it is important to choose a period during which field conditions closely resemble the behavior that the model aims to predict.

For this exercise, the model was trained with BHP data, as there were wells that were constantly optimized with the ESP pumps, also with remedial job interventions such as fractures, therefore the BHP data incorporation was valuable for the model to understand specific pressure drawdowns and skin removal with the calibration of the pressure data. The results of the simulation showed a good historical match in terms of fluid and oil production as shown in the Figure 6.

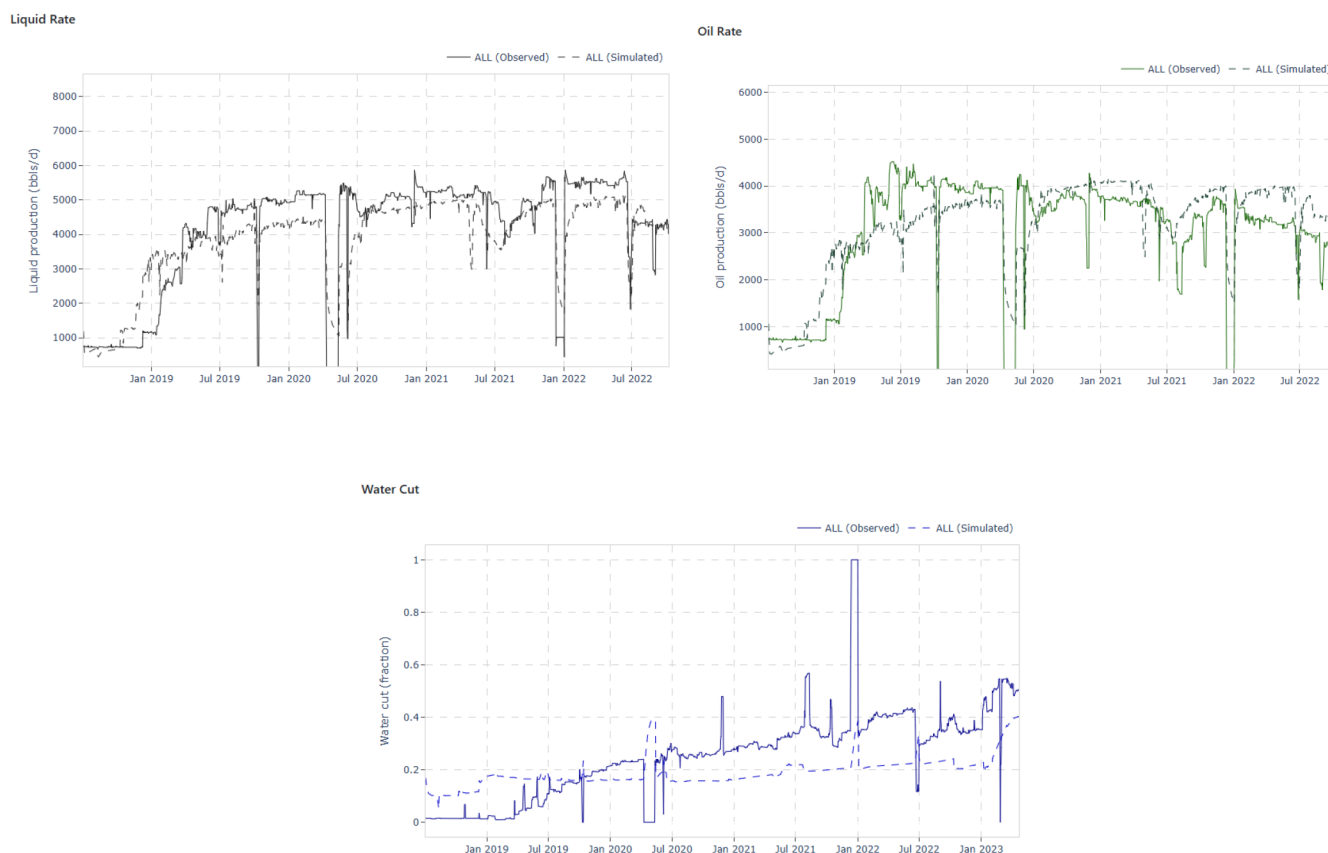


Figure 6—Historical matching of fluid and oil production

Insights & Key Findings. Figure 7 shows the injector-producer connectivity determined from fitting historical production and injection data during the period from 2018 to 2023. Figure 8 shows the sum

of the injection fractions for each injector. If there is a perfect balance of injection and production in a pattern, then the allocation fractions should sum to 1. However, if this sum is greater than 1 then it indicates that the optimizer is seeking additional influx to match the sum of the liquid production rate from offset connected wells. The offset wells are determined by those producers whose production rate signature shows maximum correlation with the injectors' injection rate signal. This additional influx is indicative of possible aquifer influx, injector outside the modelled system or expansion fluxes due to pressure contrast with the external system. Precisely defining this influx source requires other information. Similarly, if the sum is less than 1 then injection is lost to the external system or an unmodelled producer. Again, precisely defining the sink requires more information. Comparing the allocation factor information with the legacy pattern shown in Figure 4 it can be observed that the AWI solution is pointing towards possible strong influence of injectors with a few second line of producers. Correlating allocation factors with observed water cut figures highlighted producers which could be experiencing water recycling due to channeling or reaching maturity in terms of recovery, In figure 7 show how two wells had high water cut values, in the order of 90 to 95%, which were indicative that injection water recirculation could be happening, specifically wells ACAL-204 & ACAP-179, also as a product of this water injection recycling and low injection efficiency the model also shows insights of injection water losses from both injectors and poor waterflooding balancing. The fitting of water cut data using the Koval model throws insight into flooded pore volume and level of heterogeneity. This information is used to decide on the type of remedial action to be taken.

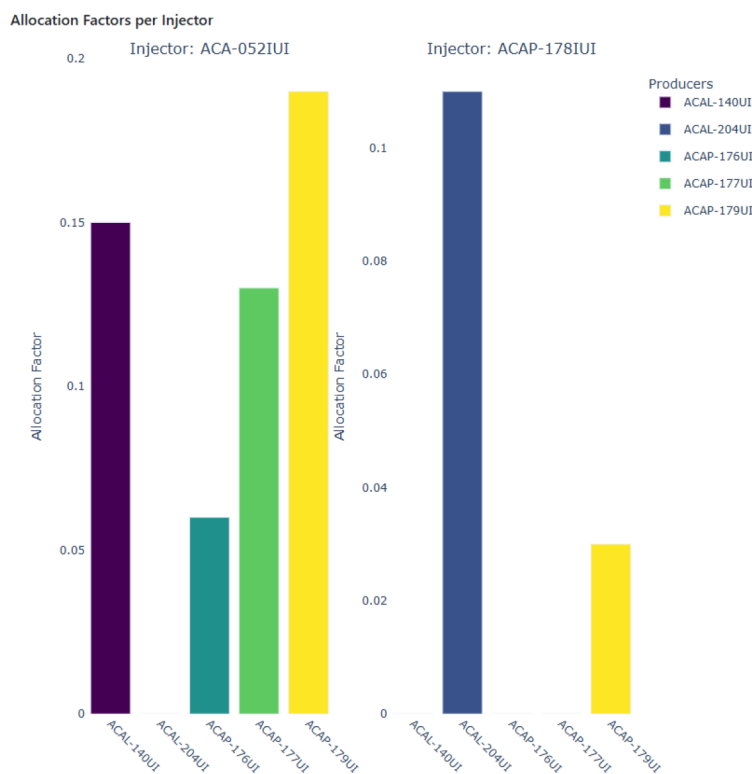


Figure 7—Injector-producer connectivity Study 1

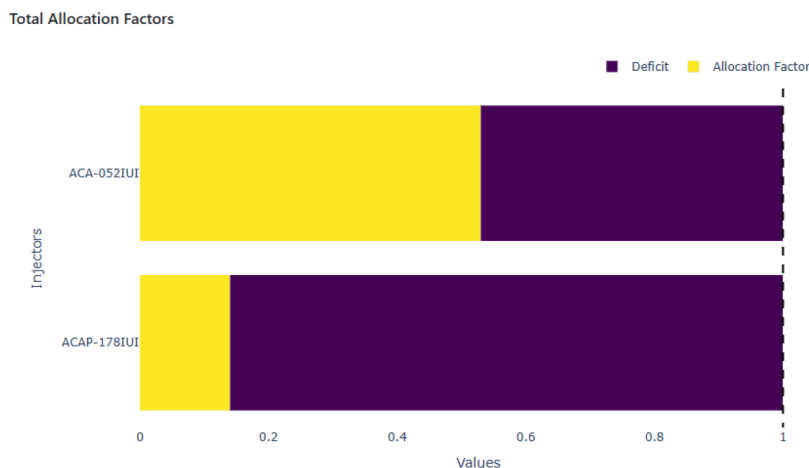


Figure 8—Modeled total allocation per injector well for Study 1

Pattern Balancing. The PFM feature was used to with the water recycling reducing strategy option for rebalancing the waterflooding pattern and enhancing oil recovery. The results, shown below, indicated that two of the highest water-producing wells—both with elevated water injection allocation factors—needed to be shut in. This recommendation aimed to redistribute the injected water streamlines toward other producing wells with more favorable response potential, ultimately improving overall sweep efficiency and oil recovery. Additionally, the optimizer suggested adjusting injection rates to achieve a more balanced distribution between the two patterns, reducing inefficiencies and enhancing reservoir performance.

The recommended actions were implemented in the field and as shown in Figure 9, the other producing wells had an almost immediate reaction with increased fluid and oil production. As a summary, the main recommended actions involved shutting down high water cut wells, increasing production in other affected wells to redistribute streamlines and adjust injection rates of the injectors to improve injection efficiency. In one well it was necessary to do a workover job to upsize the ESP pump to reach the offtake rate recommended by PFM. The strategy suggested by the workflow and PFM was successful in the field since it mitigated the production decline trend, increased production and oil recovery of the area of influence of the waterflooding patterns.

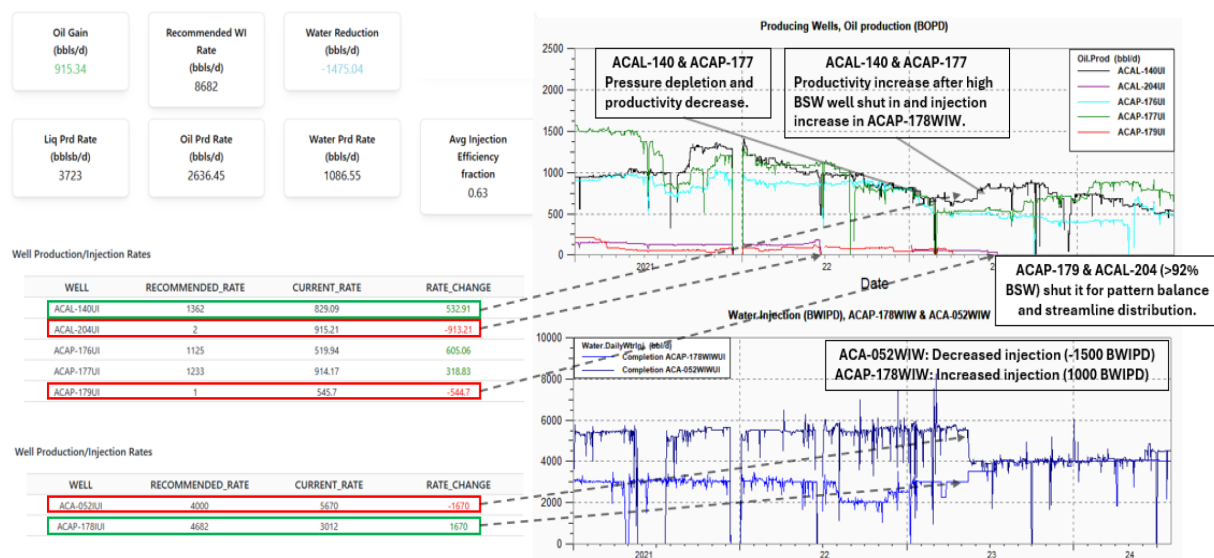


Figure 9—Results of PFM analysis and subsequent validation

Study 2

Problem Statement. This case corresponds to the Auca field, Upper T reservoir. In this second case study, a rapid evaluation was required to understand the behavior of the reservoir and take timely corrective actions due to a sharp decline in production. At that moment, it was initially assumed that injector well ACAC-079 provided the main support to producer well ACAR-132, which was experiencing a significant drop in output.

However, upon applying AWI to determine the actual contribution from the injector wells, it was revealed that injector well ACAR-128 was, in fact, the primary source of support. This new understanding of injection-production dynamics allowed for improved reservoir characterization and the implementation of optimization strategies to enhance oil production. Further details are provided below.

The area is characterized by a combination of injection patterns involving both deviated and horizontal wells, deployed in a setting with complex geology. These geological complexities such as lateral facies changes, compartmentalization, and permeability contrasts further complicate the interpretation of injector-producer connectivity and the diagnosis of production anomalies. The coexistence of different well types within the same pattern introduces additional uncertainty, as flow dynamics can vary significantly depending on the geometry and position of each well. This makes it essential to validate assumptions with data-driven tools to guide operational decisions and improve reservoir management (Figure 10).

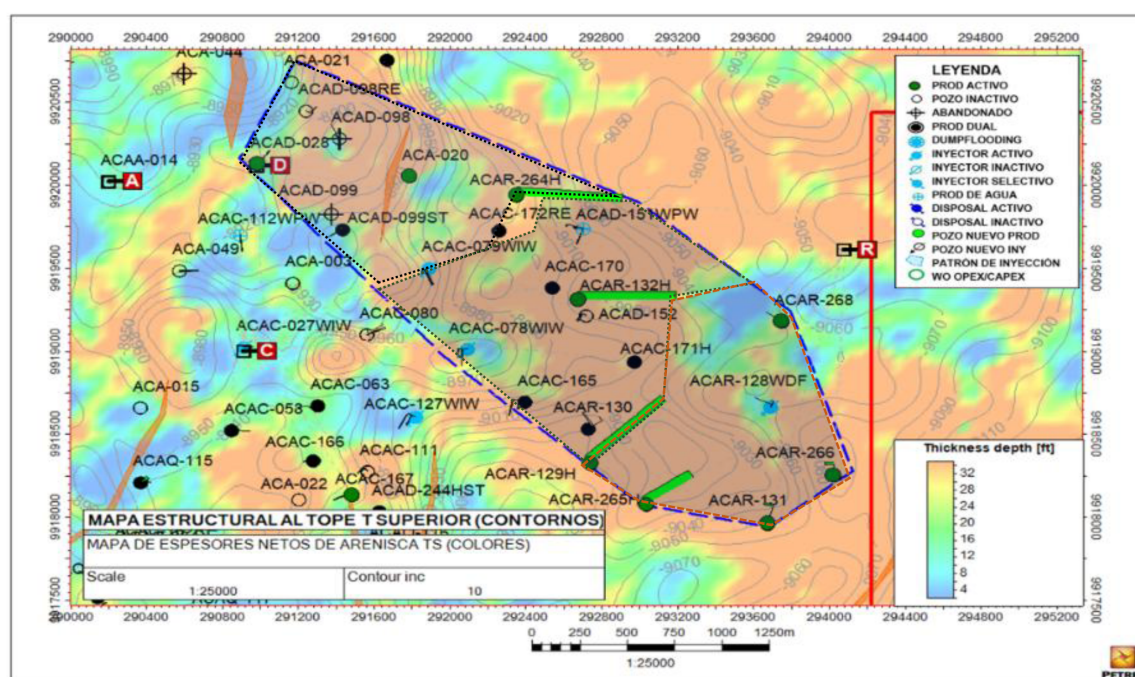


Figure 10—Location map of TS waterflooding project

Modeling. Like Case Study 1, the dataset was prepared using well production information, which is automatically ingested from the main database. For this case, the selected configuration included the Upper T Sand in the Auca field and the pads of interest: Pads R, C, D, and single-well locations. Continuing with the workflow, an automated data quality check was performed.

For model training, the entire injection project period—from October 2022 to March 2024—was selected. In this field, there were three injector wells and ten associated producer wells, with water injections starting in 2022. The selection of the training period is a critical step, as it ensures the model is trained under the same conditions in which the analysis is intended to be performed. Since the model uses daily data for training, the training period required is generally low. In some case studies, the model has been successfully

trained using as little as three months of historical data. However, in this case, a longer period was selected to ensure a comprehensive understanding of the entire analysis time.

In this case, BHP data was not incorporated, as it remained relatively constant during the training period. Additionally, the field data was considered highly reliable due to the presence of multiphase flow meters (Vx) at the station.

Once the model was trained, a good history match was achieved both at the well level and at the field level and in Liquid and oil production, as can be observed in the graph presented below (Figure 11 and Figure 12).



Figure 11—Model training results for Study Case 2 – "Field Profiles"

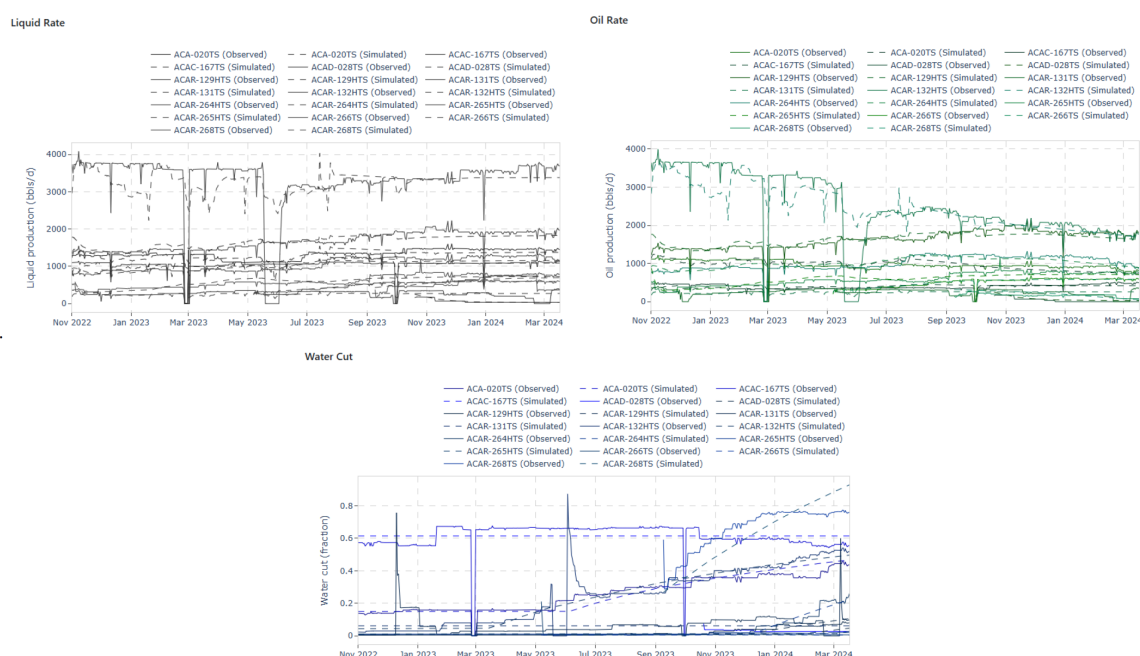


Figure 12—Model training results for Study Case 2 – "Well by well profiles"

Insights & Key Findings. Once the model is calibrated, the next step is to analyze the insights provided by the application. Key parameters can be identified, such as the connectivity between injectors and producers, allocation factors, the response time of producers to an injection pulse, the pore volume and reservoir heterogeneity reflected by the Koval factor. These outputs provide a valuable understanding of the injection-production dynamics and can be visualized in the following figures.

The most important insight provided by the application was the identification that ACAR-132H was being influenced primarily by a different injector than previously assumed. This understanding enabled the implementation of corrective actions, which were subsequently suggested by the PFM optimizer, as explained in the following section (Figure 13 and Figure 14).



Figure 13—Model training results for Study Case 2 insights – "Well-by-well profiles"

Koval Model Insights

Pore Volume per producer

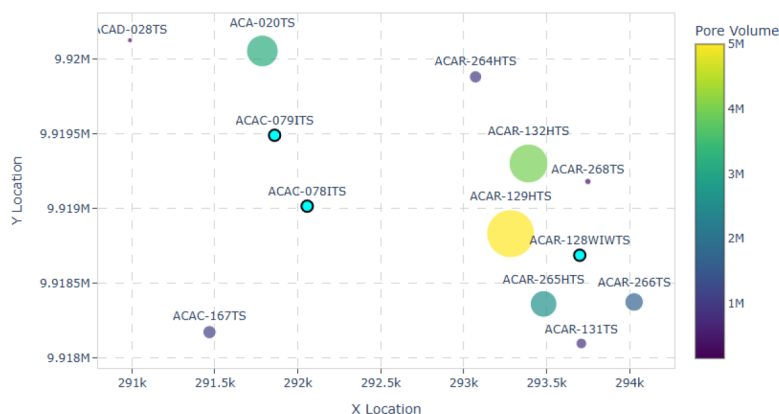


Figure 14—Koval model training results for Study Case 2 insights – "Well-by-well profiles"

Pattern Balancing. Using water recycling reduction strategy, the PFM optimizer was run to determine whether improving the injectivity of well ACAR-128 could enhance the overall performance of the injection pattern. Initially, the optimizer suggested increasing water injection by 128 barrels per day in ACAR-128. However, this was decided to try some what-if scenarios and evaluate the impact of different injection rates ranging from 150 to over 1,500 barrels of water injected per day (BWIPD). PFM also suggested reducing the injection rate in well ACAC-079, which initially contradicted the previous reservoir understanding that indicated increasing injection was the most appropriate action (Figure 15).

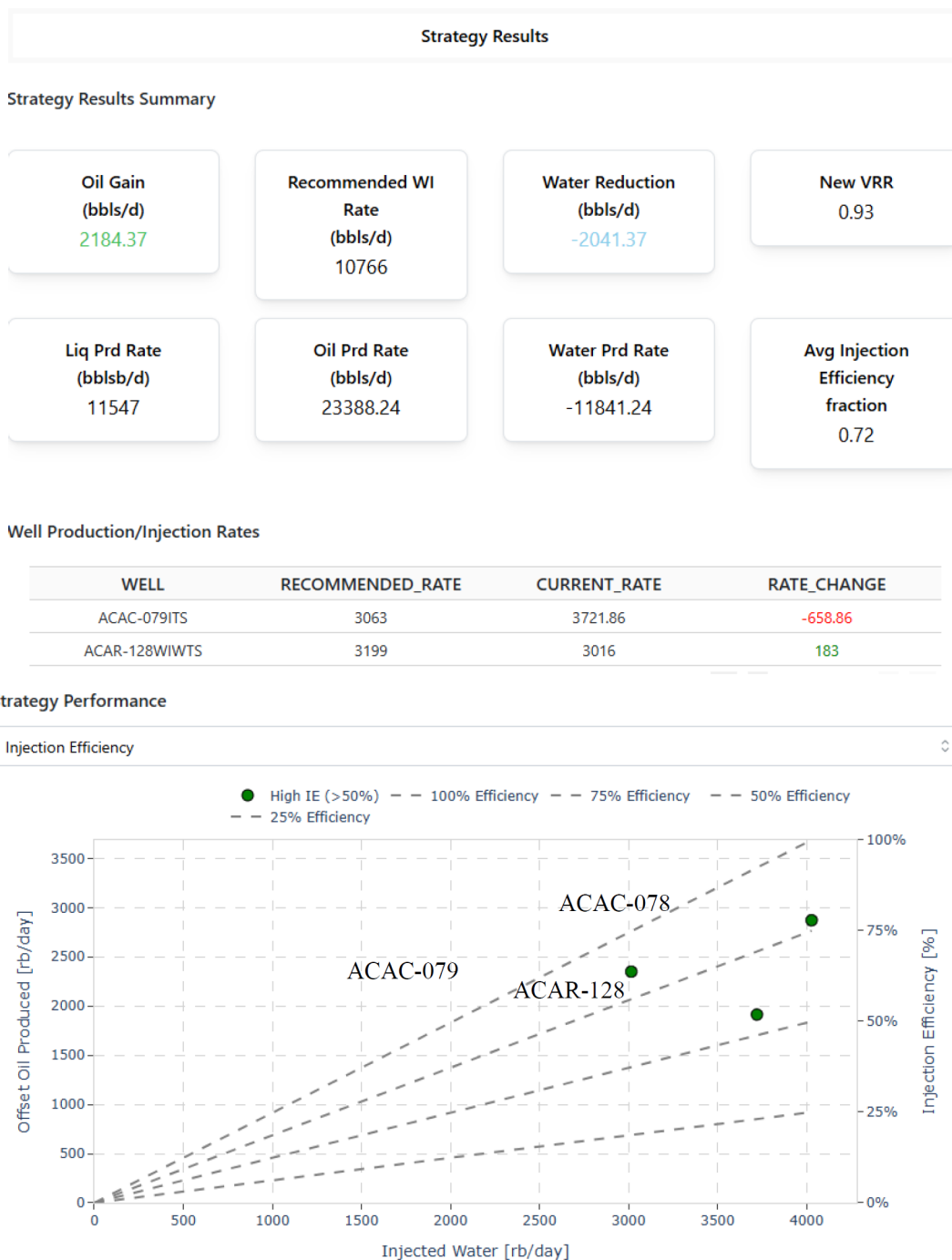


Figure 15—Results of PFM analysis case of study 2

During the forecasting phase, several scenarios were evaluated ranging from those suggested by the PFM optimizer to additional "what-if" cases. All scenarios consistently indicated that increasing injection in well ACAR-128 would lead to improvements in oil production. Based on this analysis, a decision was made to increase injection in ACAR-128 by +1,500 barrels of water per day. To enable this, a matrix stimulation was conducted on the injector. Several months later, a field evaluation confirmed the positive impact of the decision on overall field performance.

Once the stimulation was completed, the injection rate could be increased. This process took a couple of months, during which production continued to decline, particularly in April 2024. Eventually, the injection rate was successfully increased, reaching up to 5,000 barrels of water injected per day.

AWI Blind Test Validation. The same model, trained up to March 2024, was then used to forecast what the production behavior would have been, now incorporating the actual injection profile—both the loss of injectivity and the subsequent increase. The forecasted production was then compared with the actual production data recorded until November 2024.

The results showed a strong match between the forecast and actual production. The producing wells increased their output as predicted by the model, showing good alignment both in trend and magnitude. This outcome validated the results provided by the application and confirmed its reliability for supporting decision-making in the field (Figure 17).

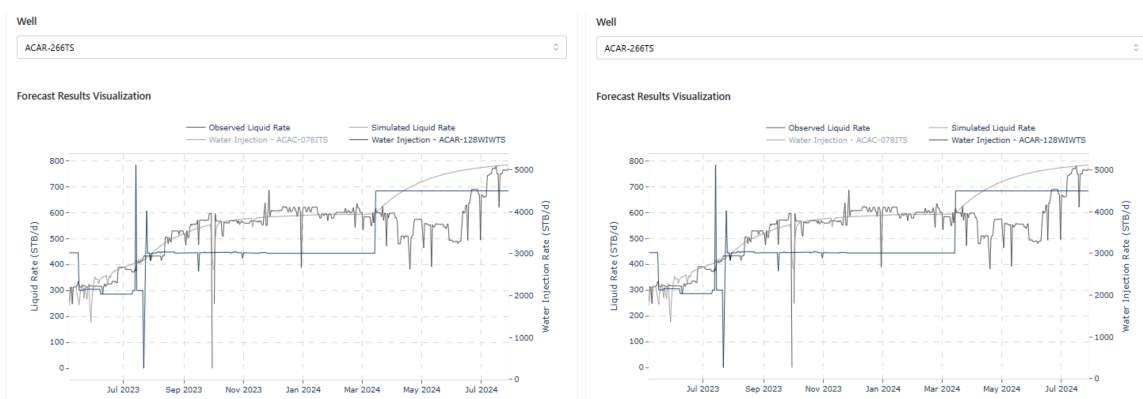


Figure 16—Results of forecast what if scenarios case of study 2

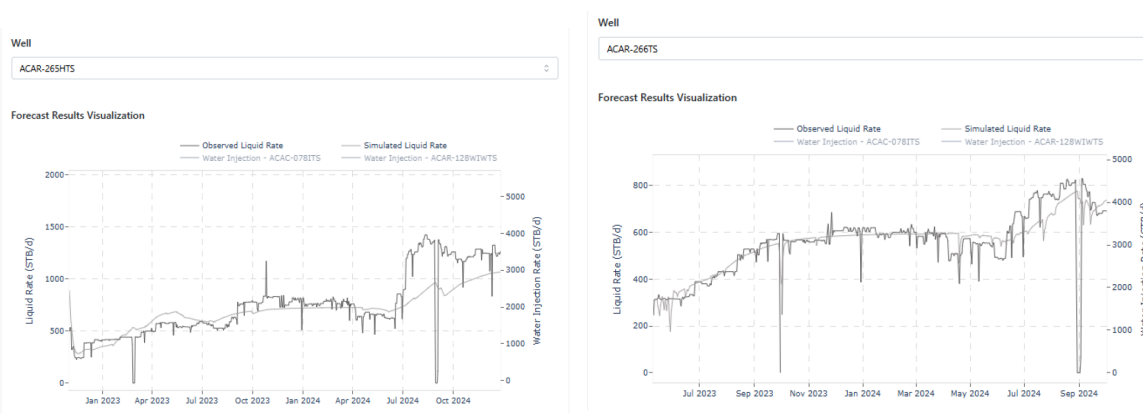


Figure 17—Results of forecast validation case of study 2

Conclusions

- In example 1 the pattern balancing strategy suggested by the application proved highly effective, helping to reverse the production decline trend while increasing both production and oil recovery within the affected area of influence of the waterflooding patterns.
- In example 2 the case proved that, even in challenging settings with heterogeneous geology and mixed well architectures, the application was able to support targeted analysis and optimization to yield actionable insights and tangible production gains.
- The automation of processes introduced into the deployable solution has been a key enabler in enhancing the efficiency of technical personnel responsible for data analysis. The implementation of direct connections to the corporate database has streamlined and secured data loading, while automated quality check routines and embedded denoising techniques have significantly improved the integrity of the analytical inputs. These advancements have led to a large upgrade of the application, increasing their robustness and reliability. Most notably, the integration of these automation features with the advanced calculation algorithms have drastically reduced analysis time from approximately 23 hours to just 5 hours being a breakthrough in efficiency and enabling faster, data-driven operational decisions.
- Since its deployment in September 2023, the application has directly contributed to over 128 KBO of cumulative oil production across the analyzed fields, generating an economic impact of 3.81 MMUSD to date.
- Beyond operational and economic benefits, the application has fostered improved cross-functional collaboration, enabling faster and more consistent responses to dynamic field conditions. Additionally, its implementation supports corporate sustainability goals, reducing the need for on-site monitoring and thereby cutting CO₂ emissions by up to 0.6 tons per month.
- The adaptation of capacitance-resistance physics-based algorithms, enhanced with machine learning components, the integration of multiple modeling approaches, and the customization to the specific reservoir conditions in Block 61 have enabled the development of a robust application for understanding and optimizing injection projects. This hybrid approach has improved the characterization of injection–production dynamics and has helped more effective decision-making for waterflood performance and optimization.

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